

# Industrial Fault Detection System Report

## Executive Summary

Analysis of 3 fault detection systems completed. 3/3 systems are ready for production deployment.  
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## Performance Summary

| Dataset        | Accuracy | F1-Score | CV Score    | Status                | Risk   |
|----------------|----------|----------|-------------|-----------------------|--------|
| CIA            | 0.920    | 0.907    | 0.985±0.009 | READY FOR PRODUCTION  | LOW    |
| BATTERY_MULTI  | 0.896    | 0.896    | 0.926±0.027 | READY WITH MONITORING | MEDIUM |
| BATTERY_SIMPLE | 0.855    | 0.855    | 0.822±0.045 | READY WITH MONITORING | MEDIUM |

### CIA Dataset Analysis

**Dataset Information:**

- Samples: 500
- Features: 10
- Ensemble Models: rf\_ultra, gb\_ultra, et\_ultra

**Performance Metrics:**

- Accuracy: 0.9200
- F1-Score: 0.9072
- Precision: 0.8948
- Recall: 0.9200
- Cross-Validation: 0.9848 ± 0.0088

**Deployment Recommendation:**

Status: READY FOR PRODUCTION  
Risk Level: LOW

### BATTERY\_MULTI Dataset Analysis

**Dataset Information:**

- Samples: 1152
- Features: 4
- Ensemble Models: rf\_ultra, gb\_ultra, et\_ultra

**Performance Metrics:**

- Accuracy: 0.8960
- F1-Score: 0.8958

- Precision: 0.8960
- Recall: 0.8960
- Cross-Validation:  $0.9264 \pm 0.0273$

**Deployment Recommendation:**

Status: READY WITH MONITORING

Risk Level: MEDIUM

***BATTERY\_SIMPLE Dataset Analysis***

**Dataset Information:**

- Samples: 728
- Features: 4
- Ensemble Models: rf\_ultra, gb\_ultra, et\_ultra

**Performance Metrics:**

- Accuracy: 0.8545
- F1-Score: 0.8545
- Precision: 0.8545
- Recall: 0.8545
- Cross-Validation:  $0.8222 \pm 0.0453$

**Deployment Recommendation:**

Status: READY WITH MONITORING

Risk Level: MEDIUM